

Group Design Project: Assignment for Proposals**Assigned: Monday, February 9, 2026****Due: Friday, February 27, 2026 via Canvas**

Your current assignment for your Group Design Project is to prepare a proposal providing background on a strategy for power generation and cooling required for a new data center. With rising AI expansion data center demand has exploded. You are tasked to work in teams which are tasked with providing the required cooling and power requirements for the assigned data center size and location. These two aspects of data centers are intertwined so you will need to communicate across subgroups to ensure you are meeting the overall demands.

Power subgroup: This is based on your assigned sustainable/renewable energy source and comparison to your assigned existing conventional technology. Power combines the actual power needs of the servers themselves along with power for any cooling demands

Cooling subgroup: This is based on the requirements of the servers within the data center. You may want to consider seasonal temperature variations in the chosen location as well as current and future chips. You will compare both air and liquid cooling technologies.

Content

Your proposal should communicate the following in coordination with both subgroups (~2 pages):

- 1) The motivation for your project.
- 2) The electrical power demand (in kW, MW, or GW) you seek to satisfy, based on the assigned scale and location for your project, including how you identified/estimated this demand. For example, identify an amount of energy consumption (frequently reported in kilowatt-hours by electricity providers for a specified time period) and divide by the time period over which it is reported.
- 3) Identification of constraints for development of your assigned sustainable/renewable technology. In particular, you should discuss as possible the following types of concerns:
 - Economic (*e.g.*, what are the current relative costs for power generated from your conventional vs. renewable sources? How about for air vs. liquid cooling?))
 - Environmental (*e.g.*, what impact does/will your assigned renewable technology and cooling approach have on the environment, including local flora and fauna?)
 - Political (*e.g.*, what regulatory guidelines, if any, exist, are there recognized political champions for your assigned renewable technology, and would the local population support your technology?)
 - Health & safety (*e.g.*, what special public health and safety concerns are associated with your assigned renewable technology?)
 - Reliability and sustainability (*e.g.*, is there an established track record for your assigned renewable technology and cooling approaches, including at the scale for your project?)Your discussion for this section should focus on concerns that are general to your technology, with comments on specifics affecting the scale of your project as relevant.
- 4) Proximity between power hub and data center
- 5) Any additional questions and/or items that you have identified that you will need to address for your project. Strong submissions identify a minimum of five follow-up questions.

Power group specific aspects (~1-1.5 pages):

- 6) A discussion of the relative advantages and disadvantages of your assigned conventional technology based on fossil/nuclear fuels
- 7) Background on your assigned sustainable/renewable technology, including historical developments and current state-of-the-art.
- 8) A discussion of the relative advantages and disadvantages of your assigned sustainable/renewable technology.
- 9) Preliminary specifications for implementing your assigned sustainable/renewable technology in your assigned project location. In particular, different types of specifications will be needed for different technologies.
 - Geothermal: specify the temperature, pressure, and phase(s) of steam from your geothermal source – These you can look up and should not need to be calculated.
 - Solar thermal: specify the design and operating temperature for your solar collector.
 - Ocean thermal: specify the temperatures and depths for your hot source and cold sink
 - Biomass/biofuel: specify the type and sourcing of your biofuel and its energy content

Cooling group specific aspects (~1-1.5 pages):

- 10) A discussion of the relative advantages and disadvantages of air and liquid cooling technologies.
- 11) Background on air and liquid approaches of server cooling including historical developments and current state-of-the-art.
- 12) Preliminary specifications for implementing your cooling technology including resources required including power or otherwise
- 13) Any design approaches that could allow for power savings

Guidelines for Submission

- 1) Each team should submit a single PDF document.
- 2) Your proposal should be a **maximum of five pages** in length (not including the cover page, references, and appendices), single-spaced and with 12-pt font (Times New Roman is preferable), 1" margins on all sides, and page numbers at the bottom of each page after the first page.
- 3) Your cover page should include your project name, the names of each team member, the course name, date, and a single paragraph summarizing what your team seeks to accomplish, including what technologies you will be comparing, where your project will be located (be creative and pick a town or specific location that seems practical and interesting!), and what electrical power demand you will deliver.
- 4) Use appendices for any supplemental information that is not absolutely critical but may be useful to the interested reader. Strong submissions typically have material provided in an appendix!
- 5) A list of references used should be placed after the body of the text and before any appendix(ces). Make sure to cite these references in the text! Minimize the use of web sites as references; stronger submissions have a minimum of three references that are not web sites. Use the following examples for formatting for references; note that a journal publication or conference proceeding should not be cited with a URL.

Journal Publications

Fulem, M., Ruzicka, K., Ruzicka, V., Simecek, T., Hulicius, E., Pangrac, J., Becker, J.,

Koch, J, and Salzmann, A. “*Vapor Pressure of Di-tert-butylsilane.*” **Journal of Chemical and Engineering Data**, Vol. 50, 2005, pp. 1613-1615.

Conference Proceedings

Zeid, I., Gupta, S.M. and Bardasz, T. “*An Analogical Problem Solving Approach to Planning for Disassembly.*” **Design for Manufacturing Assembly, Proceedings of the 1996 ASME Winter Annual Meeting**, DE-Vol. 89, 1996, pp. 79-89.

Books

Welty, J.R., Wicks, C.E., Wilson, R.E., and Rorrer, G.L. **Fundamentals of Momentum, Heat, and Mass Transfer 5th ed.**, John Wiley & Sons, Inc., 2008, pp. 5-10.

Web sites

Author or company, date found, direct link.

Cite references by either using a superscript with the number for a reference (with references numbered in the References section) or listing the first author and year in parentheses (with references alphabetized in the References section).

Assessment

Your reports will be evaluated based on the following criteria:

Technical content (*e.g.*, were all the requested items for content addressed and done so with appropriate detail?)

Quality of writing (*e.g.*, structure, grammar, clarity and conciseness)

In particular, please proofread your proposal prior to submission!

Please review the rubric with detailed breakdown of criteria for assessment before submitting your proposal to confirm key elements sought.

Students are welcome and encouraged to discuss their projects, their ideas, and problems encountered with the instructor prior to and after submission of proposals. Keep in mind that earlier discussions often are more fruitful (and pleasant) than last-minute efforts!